

5(a)(i)	<i>mark first two answers</i> ATP ; reduced NADP ; A NADPH	2
5(a)(ii)	<i>any two from:</i> 1 an accessory pigment ; 2 absorb, light (energy) / photons ; A harvest I trap / capture 3 pass on <u>energy</u> to, primary pigment / chlorophyll a / reaction centre ; 4 AVP ; e.g. <i>idea of</i> extending the range of light wavelengths absorbed / absorbs wavelengths not absorbed by primary pigment	2

Question	Answer	Marks
5(b)	100 ;	1
5(c)(i)	1 0 – 400 ; A up to 400 I units any two from: 2 temperature / <u>concentration</u> of carbon dioxide, becomes the limiting factor ; 3 (at a lower than optimum temperature) less kinetic energy so fewer collisions between rubisco and CO₂ or a lower temp will limit the rate of (named) enzyme-controlled reactions ; 4 (with a lower than optimum CO₂ concentration) less fixation of CO₂ / AW ; 5 all, enzymes / processes, in photosynthesis, already at, highest rate / optimum or maximum amount of light is being absorbed (by the pigments) ;	3
5(c)(ii)	any three from: 1 different wavelengths of light absorbed by different pigments ; 2 red light is absorbed the most ; 3 green light is, absorbed less / reflected ; 4 shorter the wavelength the greater the energy / red light has more energy (than blue light) ; ora 5 the more, light / energy, absorbed the more, light dependent reaction / photophosphorylation, occurs ; ora	3